

# Python If-Else Statements: From Basic to Advanced

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## [00:00] Introduction & Agenda

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*[Visual: You on camera or an animated intro slide with the topic and agenda]*

**Host:** "Hello everyone, and welcome back to the channel! Today, we are diving deep into one of the most fundamental concepts in Python programming: Decision Making and Control Flow using `if`, `elif`, and `else` statements.

In our daily lives, we make decisions based on certain conditions—like, 'If it is raining, I will take an umbrella; otherwise, I won't.' Python does the exact same thing.

In this complete tutorial, we will cover everything from basic `if/else` blocks to logical operators, nested conditions, `elif` ladders, and the Pythonic Ternary operator. But we aren't just going to learn theory. I have packed this session with real-world coding problems. I'll teach a concept, give you a problem, ask you to pause and code along, and then we will solve it together.

Finally, stick around until the end for our 'Predict the Output' section, where I'll expose the most common traps and bugs that trip up beginners. Let's get started!"

## [02:00] Section 1: Basic If / Else & Comparison Operators

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*[Visual: Screen recording of VS Code / Pycharm]*

**Host:** "Let's start with the basics. An `if` statement tells Python to execute a block of code only if a certain condition is `True`. If it's `False`, we can use an `else` statement to run a different block of code.

Here is the syntax:

```
if condition:
    # Code to run if true
else:
    # Code to run if false
```

Notice the indentation—the spaces before the code inside the block. Unlike C or Java that use curly braces {}, Python relies entirely on indentation to know what code belongs to the if block.

Let's jump straight into our first practice problem."

## Practice Problem 1: Even or Odd

**Host:** "Here is your task: Write a program that takes a positive integer as input and determines if it is even or odd. Hint: Use the modulus operator % which returns the remainder of a division. Pause the video, try it yourself, and resume when you're ready."

*[Visual: 3-second countdown timer graphic]*

**Host:** "Alright, let's solve it! A number is even if it is completely divisible by 2, meaning the remainder is zero."

```
number = int(input("Enter a number: "))
if number % 2 == 0:
    print("It is an Even number")
else:
    print("It is an Odd number")
```

## Practice Problem 2: Absolute Value

**Host:** "Great job. Let's try another one. I want you to print the absolute value of an inputted integer without using built-in math functions. For example, if I enter -7, the output should be 7. If I enter 5, it stays 5. Pause and try it."

*[Visual: 3-second countdown timer graphic]*

**Host:** "Here's the logic: If the number is already greater than or equal to zero, we do nothing. If it is less than zero, we multiply it by -1 to make it positive."

```
n = int(input("Enter a number: "))
if n < 0:
    n = n * -1
print("Absolute value is:", n)
```

### Practice Problem 3: Profit or Loss

**Host:** "Let's apply this to a business scenario. Take a `cost_price` and a `selling_price` as input. Determine if the seller made a profit, a loss, or broke even. Pause the video and give it a shot."

*[Visual: 3-second countdown timer graphic]*

**Host:** "Let's check our logic. If the selling price is higher than the cost price, it's a profit. If it's lower, it's a loss."

```
cp = float(input("Enter Cost Price: "))
sp = float(input("Enter Selling Price: "))

if sp > cp:
    print("You made a Profit!")
else:
    if cp > sp:
        print("You incurred a Loss.")
    else:
        print("You broke even.")
```

**Host:** "Notice how I placed an if/else inside another else? This is called nesting, which we will explore deeply very soon."

## [08:30] Section 2: Logical Operators (and, or, not)

*[Visual: Slide showing and (both true), or (either true), not (reverse)]*

**Host:** "What if we need to check multiple conditions at the same time? In C, you might use `&&` or `||`. In Python, we use plain English words: `and`, `or`, and `not`.

Let's use these in some problems."

## Practice Problem 4: Three-Digit Number

**Host:** "Write a program to check if an inputted positive integer is a three-digit number. Pause and try it!"

*[Visual: 3-second countdown timer graphic]*

**Host:** "A three-digit number must be greater than 99 AND less than 1000. So we use the and operator."

```
num = int(input("Enter a number: "))
if num > 99 and num < 1000:
    print("It is a 3-digit number.")
else:
    print("It is NOT a 3-digit number.")
```

## Practice Problem 5: Divisibility by 5 and 3

**Host:** "Next challenge: Check if a number is divisible by BOTH 5 and 3. Pause and code."

*[Visual: 3-second countdown timer graphic]*

**Host:** "For this, the remainder for both divisions must be zero."

```
num = int(input("Enter a number: "))
if num % 5 == 0 and num % 3 == 0:
    print("Divisible by both 5 and 3.")
else:
    print("Not divisible by both.")
```

**(Host Pro-tip):** "Mathematically, if a number is divisible by 5 and 3, it's divisible by 15. So `if num % 15 == 0`: would also work perfectly!"

## Practice Problem 6: Valid Triangle

**Host:** "Let's do some geometry. Take 3 sides of a triangle as input. Determine if they can form a valid triangle. The rule is: The sum of any two sides must be strictly greater than the third side. Pause and try!"

*[Visual: 3-second countdown timer graphic]*

**Host:** "We need to check three conditions, and ALL of them must be true. So we will chain them with and."

```
a = int(input("Side A: "))
b = int(input("Side B: "))
c = int(input("Side C: "))

if (a + b > c) and (b + c > a) and (c + a > b):
    print("Valid Triangle!")
else:
    print("Invalid Triangle.")
```

## [14:00] Section 3: Nested if-else Blocks

*[Visual: Diagram showing an if block inside another if block]*

**Host:** "Sometimes, you need to check a condition only after a previous condition has proven to be true. This is called Nesting. Let's look at a classic problem without using logical operators."

### Practice Problem 7: Greatest of 3 (Nested Logic)

**Host:** "Find the greatest of 3 numbers (a, b, c) without using the and operator. You must rely solely on nested conditional blocks. Pause and think this through."

*[Visual: 3-second countdown timer graphic]*

**Host:** "Here is how we think about it: If a is greater than b, we then step inside and check if a is also greater than c."

```
a, b, c = 10, 25, 15

if a > b:
    if a > c:
        print("A is the greatest")
    else:
        print("C is the greatest")
else:
    if b > c:
        print("B is the greatest")
    else:
        print("C is the greatest")
```

## [18:00] Section 4: Elif Ladders (Multiple Conditions)

*[Visual: Code comparing heavily nested if/elses vs a clean elif ladder]*

**Host:** "Nesting too deep makes our code messy and hard to read. When we have multiple mutually exclusive conditions, Python gives us the `elif` keyword, which stands for 'else if'."

### Practice Problem 8: Grading System

**Host:** "Take a student's percentage as input. If it's > 80, print 'A'. > 60 print 'B'. > 40 print 'C'. Otherwise, print 'D'. Pause and try it using `elif`."

*[Visual: 3-second countdown timer graphic]*

**Host:** "Let's see how clean this looks with an `elif` ladder."

```
marks = int(input("Enter marks: "))

if marks > 80:
    print("Grade: A")
elif marks > 60:
    print("Grade: B")
elif marks > 40:
    print("Grade: C")
else:
    print("Grade: D")
```

**Host:** "Notice that order matters! Python checks from top to bottom. Once a condition is true, it skips the rest of the ladder."

## Practice Problem 9: Point Location

**Host:** "Given an (x, y) coordinate, determine if it lies on the x-axis, the y-axis, or exactly at the origin (0,0). Pause and give it a go."

*[Visual: 3-second countdown timer graphic]*

**Host:** "Let's solve it:"

```
x = int(input("Enter X coordinate: "))
y = int(input("Enter Y coordinate: "))

if x == 0 and y == 0:
    print("Point is at the Origin.")
elif x == 0:
    print("Point lies on the Y-axis.")
elif y == 0:
    print("Point lies on the X-axis.")
else:
    print("Point is somewhere on the plane.")
```

## [22:30] Section 5: The Ternary Operator (Python's Inline If-Else)

*[Visual: Syntax text pop up: [True Value] if [Condition] else [False Value]]*

**Host:** "If you want to write cleaner, shorter code, Python has a Ternary Operator. It lets you write a simple if/else statement in a single line."

## Practice Problem 10: One-Line Even/Odd

**Host:** "Rewrite our very first Even/Odd program in just one line of code. Pause and try it."

*[Visual: 3-second countdown timer graphic]*

**Host:** "Here is the magic one-liner:"

```
num = 10
result = "Even" if num % 2 == 0 else "Odd"
print(result)
```

**Host:** "This reads just like English: Assign 'Even' if the number modulo 2 is 0, else assign 'Odd'."

## [25:00] Section 6: "The Real Thing" - Truthy vs. Falsy

*[Visual: Zoom in closely on host's face for emphasis]*

**Host:** "Now, I am going to teach you a secret about Python that separates beginners from pros. An if statement doesn't actually need a mathematical comparison like `x > 5`."

In Python, 0 is evaluated as False. Any non-zero number (like 1, -5, or 100) is evaluated as True. Empty strings and empty lists are False, while filled ones are True."

Let's look at this:

```
if 3 + 2 % 5:
    print("Will this print?")
else:
    print("Or will this?")
```

**Host:** "2 modulus 5 is 2. 3 + 2 is 5. Because 5 is a non-zero number, it evaluates to True! So the first block prints."

## [27:30] Section 7: "Predict the Output" (Common Traps & Bugs)

*[Visual: Graphic overlay saying "BUG HUNTING"]*

**Host:** "To wrap up, let's play a game of 'Predict the Output'. I'm going to show you common traps. Don't run the code—try to predict what it will do."

### Trap 1: Indentation Errors

```
if 5 > 3:
print("Hello")
```



**Host:** "In C, missing brackets might just run the next line. In Python, this throws an `IndentationError`. Python requires strict spacing!"

### Trap 2: Assignment vs. Equality

```
x = 10
if x = 5:
    print("X is 5")
```

**Host:** "In C, this assigns 5 to X and evaluates to True. But Python is smarter! It protects you from this mistake by throwing a `SyntaxError`. Always use `==` for comparison!"

### Trap 3: Type Equality

```
if 3 == 3.0:
    print("Match 1")
if "3" == 3:
    print("Match 2")
```

**Host:** "What prints? Only 'Match 1'! Python recognizes that the integer 3 and float 3.0 hold the same numeric value. However, a string '3' will NEVER equal an integer 3."

## [30:00] Conclusion & Homework

*[Visual: Screen shows the homework problems clearly formatted]*

**Host:** "You guys have done an incredible job today! We covered basic branching, logical operators, nested if-elses, elif ladders, and python's ternary operator.

But practice makes perfect. Here is your homework. Write the code for these in the comments below, and I will pin the best solutions!

1. **Leap Year Checker:** Take a year as input and check if it's a leap year. (Careful, the logic is trickier than you think!)
2. **Greatest of 4:** Take 4 positive integers and print the greatest.
3. **Smallest of 3:** Take 3 numbers and find the minimum using only nested if-else logic.
4. **Collinear Points:** Take 3 coordinate points (x1, y1), (x2, y2), (x3, y3) and calculate their slopes to find if they fall on a single straight line.

If you found this video helpful, please smash that like button, subscribe to the channel, and hit the bell icon so you don't miss our next Python masterclass. Keep coding, keep learning, and I'll see you in the next video!"

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## Complete Practice Problem Set

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The following are the core problems covered during the tutorial, categorized by the concept they enforce.

### Basic if / else Problems

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- **Even or Odd:** Determine if a given positive integer is even or odd.
- **Divisibility by 5:** Check if a given positive integer is divisible by 5.
- **Leap Year Checker:** Write a program to determine whether a given year is a leap year.
- **Absolute Value:** Print the absolute value of an inputted integer without using built-in math functions.
- **Profit or Loss:** Take cost price and selling price as input, and determine if the seller made a profit, loss, or broke even.
- **Area vs. Perimeter:** Given length and breadth, check if the area of a rectangle is greater than its perimeter.

### Logical Operators (and / or)

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- **Three-Digit Number:** Check if an inputted number is a three-digit number.
- **Divisible by Both:** Tell if a number is divisible by 5 AND 3.
- **Divisible by Either:** Tell if a number is divisible by 5 OR 3.
- **Greatest of 3 (Flat logic):** Take 3 integers and print the greatest among them using and operators.
- **Greatest of 4:** Take 4 positive integers and print the greatest.
- **Valid Triangle:** Take 3 sides as input and tell if they can form a valid triangle.
- **Youngest of 3:** Take the ages of Ram, Shyam, and Ajay, and determine who is the youngest.
- **Divisibility with Exclusions:** Tell if a number is divisible by 5 or 3, but NOT divisible by 15.

## Nested if-else Blocks & elif Ladders

- **Greatest of 3 (Nested):** Find the greatest of 3 numbers without using logical and/or operators.
- **Smallest of 3:** Find the minimum of 3 numbers using nested if-else logic.
- **Grading System:** Take a student's percentage as input and print their text grade.
- **Collinear Points:** Given 3 coordinate points, check if they fall on a single straight line.
- **Point Location:** Determine if an (x,y) coordinate lies on an axis or the origin.

## 15 Additional Scenario-Based Challenges

Test your skills with these real-world situational problems. Each scenario requires proper application of Python conditional logic to solve.

### 1. E-Commerce Discount Engine

You are building the checkout logic for an online jewelry store (like Minivy). If the cart total is over 5,000 BDT, apply a 15% discount. If it's between 2,000 BDT and 4,999 BDT, apply a 5% discount. Otherwise, no discount. Calculate and print the final payable amount.

### 2. Dhaka Traffic Delivery Routing

A delivery app charges a base fee of 60 BDT. Ask the user if their route crosses the "Kuril Flyover" during "Rush Hour" (Yes/No inputs). If both are Yes, add a 40 BDT heavy traffic surcharge. If only one is Yes, add 20 BDT. Print the final delivery fee.

### 3. Hackathon Team Eligibility

For a university event like the CSE Carnival, a team is only eligible if they have exactly 3 members, and all members are currently enrolled undergraduate students. Write a program that takes these variables as inputs and prints whether the team is approved.

#### 4. App Login Role Validation

You are building the login logic for a student app (like Grady). Take a username and role as input. If the role is "Admin", print "Full Access Granted." If the role is "Student" and their account status is "Active", print "Student Dashboard Loaded." Otherwise, print "Access Denied."

#### 5. Smart Home AC Controller

Write a script for a smart thermostat. If the room temperature is above 28°C and the room is occupied, turn the AC on to "Cool Mode". If it's below 20°C and occupied, turn on "Heat Mode". If the room is empty, turn the system "Off" regardless of temperature.

#### 6. Movie Theater Age Restriction

A cinema is showing an R-rated movie. Ask for the customer's age. If they are 18 or older, print "Ticket Issued." If they are under 18, ask if they are accompanied by an adult. If yes, issue the ticket; if no, deny entry.

#### 7. ATM Withdrawal Logic

Ask the user for their account balance and the amount they wish to withdraw. The withdrawal is only successful if: 1) The amount is a multiple of 100, and 2) The amount is less than or equal to their balance. Output the appropriate success or error message.

#### 8. Password Strength Checker

Take a password string as input. If the password is at least 8 characters long, contains a special character (like @, #, or \$), and includes numbers, print "Strong Password." If it only meets the length requirement, print "Moderate." Otherwise, print "Weak."

### 9. Game Character Health Status

In a video game, a character's maximum health is 100. Based on the current health input: print "Dead" if 0, "Critical" if 1-20, "Wounded" if 21-50, and "Healthy" if 51-100. Handle invalid inputs (like negative numbers or >100).

### 10. Flight Baggage Overweight Penalty

An airline allows 20kg of free baggage. Take the baggage weight as input. If it's under 20kg, print "No extra charge." If it's between 20.1kg and 30kg, charge a \$50 flat fee. If it exceeds 30kg, charge \$50 plus \$10 for every additional kilogram above 30.

### 11. Employee Bonus Calculator

An employee receives a bonus based on their years of service and performance rating (1 to 5). If they have been at the company for >5 years and have a rating of 4 or 5, they get a 20% bonus. If >5 years but rating is <4, they get 10%. If <5 years, they get a 5% bonus regardless of rating.

### 12. Streaming Service Feature Access

A streaming platform has three tiers: Free, Basic, and Premium. Take the user's tier as input. If Premium, print "4K resolution, No Ads, Downloads allowed." If Basic, print "1080p resolution, No Ads." If Free, print "720p resolution, Contains Ads."

### 13. Parking Garage Ticketing

A parking garage charges \$5 for the first 2 hours. After that, it charges \$2 for every additional hour, with a maximum daily cap of \$20. Write a program that takes hours parked as input and calculates the total fee.

#### 14. Blood Donation Eligibility

To donate blood, a person must be between 18 and 65 years old, weigh at least 50kg, and must not have donated blood in the last 90 days. Write a program taking these three inputs to determine if the user is eligible to donate today.

#### 15. Vending Machine Dispenser

A vending machine sells Water (\$1), Soda (\$1.50), and Juice (\$2). Ask the user for their selection and the amount of cash inserted. If the cash is exact, print "Dispensing item." If the cash is more, print "Dispensing item. Change: \$X." If the cash is less, print "Insufficient funds. Please insert \$X more."